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Reflections on Data Science 2026

Generative BS assessment

1 Design

The goal of our experiment was to evaluate how generative AI produces and supports factual claims when answering questions of varying constraints. One may ask: *How does a model behave when asked an open-ended question versus a narrowly-defined one?* To answer this question, we selected two models, ChatGPT (Instant 5.3) and Gemini (Gemini 3), and designed two prompts that differ in scope and ambiguity. We chose the above-mentioned models because they are among the most used generative AI systems as of now, making them good candidates to be evaluated for their reliability in real-life usage. By testing two models under the same conditions, we get the possibility to see variation in their claims and factual reliability.

The experimental setup consisted of two main prompts aiming to capture different types of model behavior. The first prompt, *"What is a tomato?"*, is intentionally open-ended and ambiguous. Tomatoes are classified differently depending on the domain in which they are mentioned, e.g., in botany, culinary practices, or legal cases. Based on this diverse classification, we wanted to see whether the models would focus on one domain or incorporate multiple ones in their answer. To better analyze this, each statement made by the model was labeled belonging to one of the following domains: Culinary (cooking), Botany (science), Law (history), or Other. Having these categories allowed us to evaluate how the models distribute their statements across domains with an open-ended question. Finally, by asking an open-ended question, the hypothesis is that there will be a big set of possible valid answers. We assumed that the models might favor the safest and true answers, since it has many options to choose from, thus avoiding the uncertain answers. To test this, each statement was labeled as TRUE or FALSE based on whether a relevant and credible source confirmed the claim of the model.

The second prompt, *"What historical or legal events influenced how tomatoes are classified?"*, was designed to be a more narrowly-focused question. The idea was that by restricting the scope to only be in the legal and historical domain, the number of possible valid statements would also be reduced. In other words, we hypothesized that the model would be more prone to making incorrect

statements under this constraint. Just like in the first prompt, the resulting statements were labeled as TRUE or FALSE based on supporting evidence from sources. We also analyzed the domain distribution of the statements. Since we are already asking for historical and legal events, we expected that most statements would be about law or history, but we were interested in whether the models would include anything from the other domains despite the narrow framing of the prompt.

For the data collection process, each group member queried both GPT and Gemini using the prompts, one after the other, in clean chat sessions to avoid any influence from prior context. From each response, factual claims (statements) made by the models were extracted. Each claim was labeled with its correctness, domain classification, and a supporting source, which could be a URL, "No direct source" (if source not found), or "Common sense" (if source not needed). This process resulted in a dataset containing several claims per system, per prompt, and per person, making quantitative and qualitative analysis possible. See the Appendix A as a small sample of the resulting dataset.

2 Variation in claims

In total, we got 116 statements from GPT and 135 from Gemini. Per prompt, that is 104 statements for prompt 1 and 147 statements for prompt 2. In total, we got 251 statements combined for both models and prompts.

	GPT	Gemini	Total
Prompt 1	51	53	104
Prompt 2	65	82	147
Total	116	135	251

Table 1: Total number of statements generated by GPT and Gemini models across Prompts 1 and 2.

The statements per person also varied. See the table below for the distribution of statements per person per model, per prompt.

Person	Total claims		Average claims per prompt	
	GPT	Gemini	GPT	Gemini
Christian	24	43	12.0	21.5
David	37	36	18.5	18.0
Lucie	32	33	16.0	16.5
Matus	23	23	11.5	11.5
Total	116	135	14.5	16.9

Table 2: Total and average number of claims per person and model across Prompts 1 and 2.

See the distribution of claims across persons for prompts 1 and 2, comparing GPT and Gemini

models in the Appendix A.

The process of identifying claims varied, especially when deciding whether a statement should be split into multiple smaller ones or kept together and treated as single claims. An example would be claims such as: "A tomato develops from the flower of the plant and contains seeds." Some of our group members kept it as is, so one claim, while the others split it into "A tomato develops from the flower" and "A tomato contains seeds". Therefore, there was a disagreement in the process of identifying statements, and this directly influenced the number of claims produced by each model. After comparing these approaches, we ended up with the conclusion that it is probably better not to split, as when you split, you sometimes lose context, and you can end up with a lot of duplicate claims that just artificially inflate the number of claims produced per model.

The claims were more or less the same; most responses followed a consistent structure, usually talking about the botanical and culinary facts about tomatoes at the beginning, followed by the U.S. Supreme Court case *Nix v. Hedden*, and then the early belief that tomatoes were poisonous or the other way around. This consistency shows that both models rely on a shared set of highly probable and widely available information.

However, there was still variation in less central claims. Some statements only appeared in outputs for some of us, such as references to tomatoes being considered a "superfood" ("Tomatoes are famous for being a "superfood" in the produce aisle") or more detailed cultural interpretations. These claims indicate that while the models agree on core knowledge, they diverge when generating supplementary. Also, even when the same fact was expressed, it was often phrased differently across the different responses, which just complicated the identification and comparison of claims.

In addition to the qualitative analysis, we also examined how claims are distributed across domains depending on the prompt. In the Design section, we mentioned our interest in how the level of constraint on the prompt influences the domains represented in the answers of the models. See Figure 2 (in Appendix A) showcases how prompt 1 (open-ended) led the models to generate mostly claims within the Botany (science) domain, which accounted for 57.7% of all claims. In contrast, domains such as Culinary (cooking) and Law (history) made up 16.3% each. Claims that didn't fit any domain, talking about general information or filler text, were put into the Other domain, which was 9.6% of claims. The results are interesting because one would think claims would be more evenly distributed between domains due to the ambiguity of the term tomato. Instead, the models preferred generating statements explaining the science behind tomatoes in a botanical context. This could suggest that when faced with open-ended questions, models default to more formal domains such as science. For prompt 2, the results in Figure 2 align more closely with our hypothesis. Since the prompt explicitly focuses on historical and legal events, a majority of claims (70.7%) fall within the Law (history) domain. The rest of the claims are shared across the botany (13.6%), culinary (8.2%), and other (7.5%)

domains. While this distribution is expected, it is interesting to see how the models include information from other domains regardless of the constrained framing of the prompt. Furthermore, Gemini not only produced more claims, but also longer ones on average, as shown in Table 4.

Overall, our findings show that differences in both model behavior and human annotation decisions contribute to variation in claims. The models themselves produce consistent core information; they also focus on a dominant domain depending on the prompt, regularly incorporating cross-domain details. At the same time, the process of extracting and defining claims introduces additional variability.

3 Variation in BS

All of the group members used the same procedure when searching sources for the claims (statements) made by the models. The method consisted of rephrasing the claim into a shorter version of itself in the Google Chrome browser. For example, instead of searching the whole claim "In New Jersey the tomato is classified as an official state fruit in 2003", a shortened version would be written as "New Jersey State Fruit 2003". If this approach didn't yield any relevant sources, then the claim would be rephrased into a question, "What was the official state fruit of New Jersey in 2003?" to either confirm or disprove the claim. If none of the above methods led to a credible source, the claim was labeled as having "No direct source". Additionally, if a claim made sense logically, but had no direct and supporting source, it was labeled as "Common sense". The result of this process can be seen in Figure 3 (in Appendix A), where the distribution of claims by source is shown. For 221 out of 254 claims, a direct source was found; then 25 had no direct source, and 8 claims were labeled as common sense. For the 221 claims with a source, 213 of them had a source supporting the claim, while the remaining 8 had a source contradicting the claim. Overall, Figure 3 indicates that approx. 84% of the claims were supported by a source, while some portion of the claims lacked support.

The types of credible sources used for verification were research papers, articles, web pages, legislation documents, and Wikipedia (if sources were included). Initially, we assumed that using research papers would be the most ideal form of evidence to conclude whether a claim was TRUE or FALSE. However, it was proved difficult and time-consuming to find relevant research papers, especially for general or historical information. Therefore, most of the sources in the dataset ended up being articles, web pages, legislation documents, and Wikipedia.

To further analyze the variation in factual correctness, we compared the proportion of FALSE claims between prompts 1 and 2. As previously described in our Design section, we hypothesized that the narrowly defined prompt 2 would be prone to a higher rate of incorrect claims due to the constrained set of possible answers. The data in Figure 4 (in Appendix A) shows that 15% of the

claims generated by prompt 2 were FALSE, while prompt 1 had a slightly lower proportion of 9.6% FALSE claims. This supports the hypothesis that restricting the scope of the prompt increases the likelihood of incorrect or unsupported statements made by the models.

It should be noted that for our dataset, most of the "No direct source" claims were also labeled as FALSE, since no supporting source could be found. A typical example of a FALSE claim from prompt 2 is the following: "Even after the Supreme Court ruling, the identity crisis continued at the state level, leading to some clever legislative gymnastics". Here, it was mentioned that the Supreme Court had ruled the tomato as a vegetable for the purpose of trade, and that itself is factually correct. However, there was no evidence found to support the claim of a continued state-level confusion with other legislative cases. Similar FALSE claims were observed, where the models present speculative implications as definitive, rather than expressing uncertainty.

4 Verdict

Based on our experiment, GPT and Gemini can be considered moderately reliable sources of information for well-known and widely documented simple topics. With approximately 84% of claims supported by a verifiable source and a relatively low false claim rate of 9.4% for open-ended prompts, both models perform reasonably well when the question domain is broad, and the correct answers are abundant online. However, this conclusion has important limitations in its generality. Our experiment was narrow in scope; we tested only two prompts on a single topic (tomatoes), with a dataset of 254 claims collected by 4 annotators. The topic itself is well represented across the web, meaning the models had good training to turn back to, and results would likely differ for niche or rapidly evolving subjects. Furthermore, the false claim rate increased to 14.9% for the narrower prompt, suggesting that as questions become more specific, models increasingly present speculative or unverifiable statements as fact. To draw broader conclusions, future work would need to test a wider range of topics and prompt types.

A Claim Data

As we have over 250 claims for all group members combined, we included the claim data as a table in the submission package. (claim-data.xlsx)

Person	Model	Prompt	Statement	Correct	Domain	Source
Lucie	GPT	1	A tomato is the edible fruit of the plant <i>Solanum lycopersicum</i> .	TRUE	Botany (science)	wikipedia.org
Lucie	GPT	1	<i>Solanum lycopersicum</i> is a part of the nightshade family.	TRUE	Botany (science)	Research Gate: Medicinal Plants from Solanaceae Family
Lucie	GPT	1	Tomatoes are botanically classified as fruits.	TRUE	Botany (science)	Frontiers: Tomato Fruit Development and Metabolism
Lucie	GPT	1	Tomatoes originated in South America.	TRUE	Law (history)	britannica.com
Christian	Gemini	2	The debate over whether a tomato is a fruit or a vegetable is one of the world's most famous "knowledge vs. wisdom" dilemmas.	FALSE	Other	No direct source.

Table 3: Example of labeled dataset with extracted claims (statements), correctness, domains, and sources.

B Plots

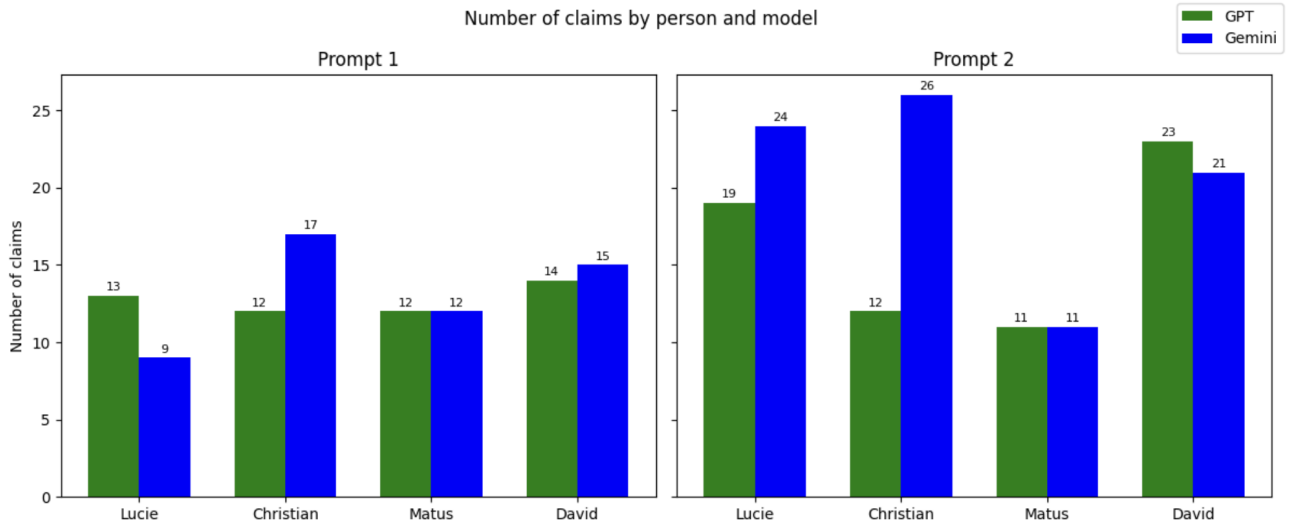


Figure 1: Distribution of claims across persons (Lucie, Christian, Matus, David) for Prompts 1 and 2, comparing GPT and Gemini models.

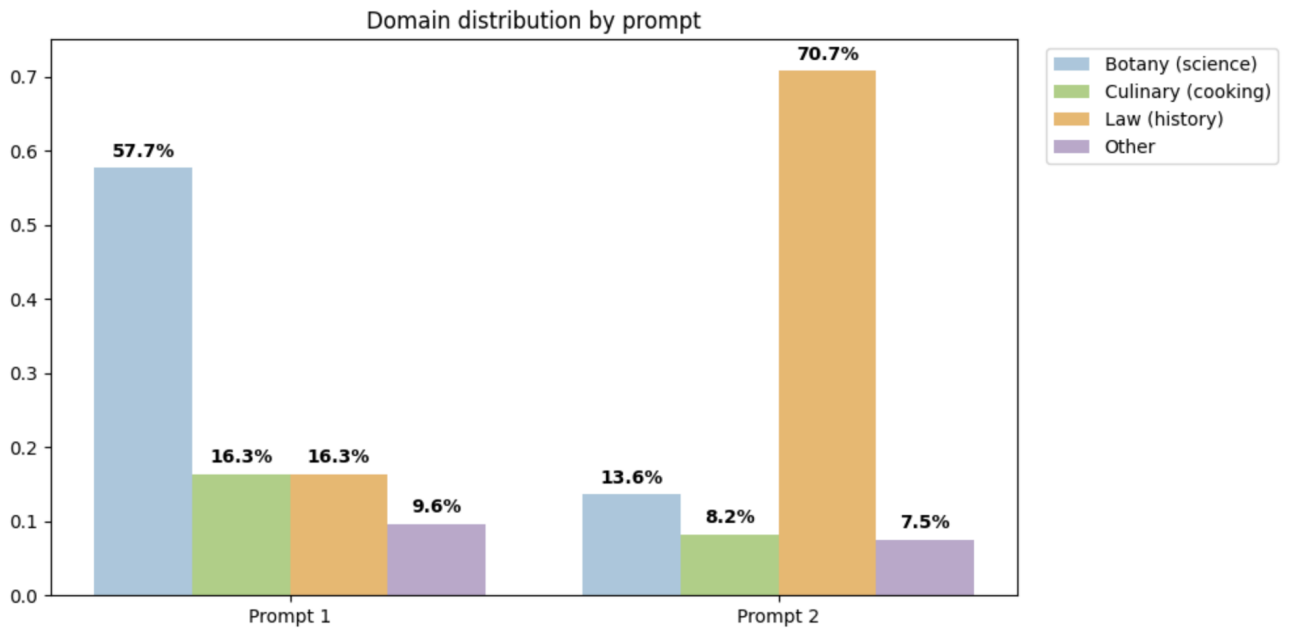


Figure 2: Normalized distribution of claims across domains for Prompt 1 (open-ended) and Prompt 2 (narrow). The figure shows how the proportion of statements varies between botany, culinary, law, and other domains depending on the level of constraint in the prompt.

Model	Average statement length (words)
GPT	10.0
Gemini	13.7

Table 4: Average number of words per statement for GPT and Gemini models.

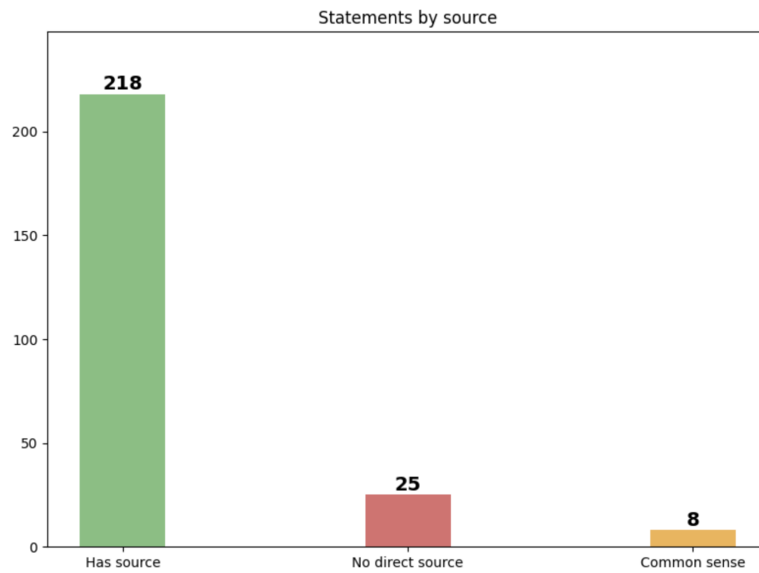


Figure 3: Distribution of statements based on source availability: statements with supporting sources (green), no direct source (red), and common sense (orange).

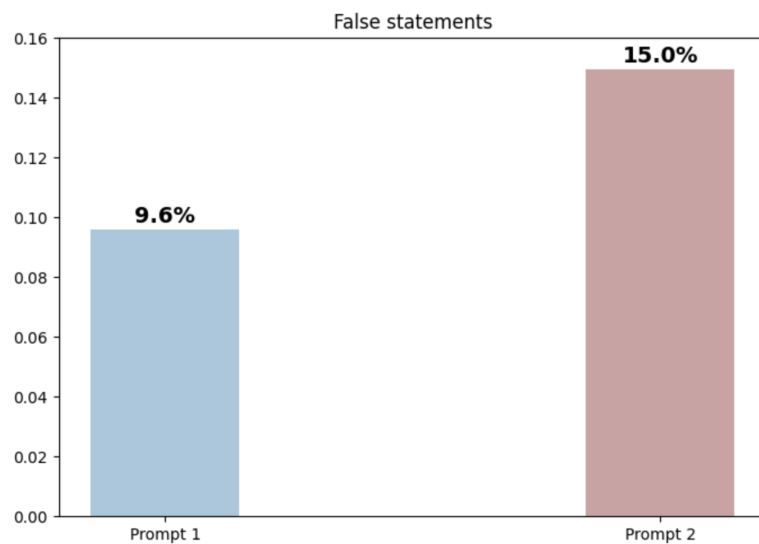


Figure 4: Prompt 1 (blue) and Prompt 2 (pink) show the proportion of FALSE statements out of all statements made by both GPT and Gemini.